

EC824A

Nonparametric Regression

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Kernel Regression

- i.i.d. sample $(Y_1, X_1), \dots, (Y_n, X_n)$, $Y_i \in \mathbb{R}$, $X_i \in \mathbb{R}^d$
- Interested in $\mathbb{E}[Y_i|X_i = x]$, *nonparametric regression*:

$$\mathbb{E}[Y_i|X_i = x] = m(x); \quad Y = m(X) + \varepsilon; \quad \mathbb{E}[\varepsilon|X = x] = 0 \quad \forall x$$

- Last “condition”: $\varepsilon = Y - m(X)$, hence $\mathbb{E}[Y - m(X)|X] = \mathbb{E}[m(X) - m(X)] = 0$
- Estimating “ $m(x)$ ” is called *nonparametric regression* or *smoothing*, or *nonparametric prediction* when prediction is the objective
- Eg 1: log-wage vs. experience (years) [Henderson and Souto (2018)]
- Eg 2: house price vs. lot size + sq. footage + bedrooms + neighbour avg. income, etc. [Yatchew (1998)]
- Eg 1 is scalar, and much easier to deal with for nonparametric regression – Eg 2 becomes less and less tractable as number of covariates, d , diverges.

Example – log-wage vs. experience

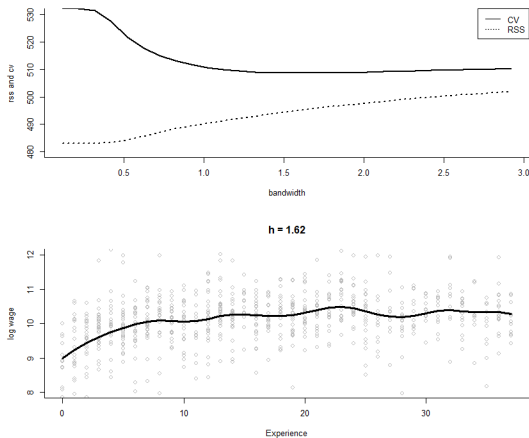


Figure: Log-wage nonparametrically regressed on experience [Henderson and Souto (2018)]

Mean Square error (pointwise)

- Pointwise MSE (aka pointwise risk) – sometimes just called *error*

$$MSE(m(x), \hat{m}(x)) = \mathbb{E}[(m(x) - \hat{m}(x))^2]$$

- Error: *empirical error* $\|m - \hat{m}\|_n^2 = n^{-1} \sum_i (m(X_i) - \hat{m}(X_i))^2$;
or L^2 -error $\|m - \hat{m}\|_2^2 = \mathbb{E}[(m(X) - \hat{m}(X))^2] = \int (m(x) - \hat{m}(x))^2 dF_X(x)$
- Subtle preference: for random X_i empirical or L^2 error can be used; for fixed X_i it only really makes sense to use empirical error (we will focus on random X_i)
- For i.i.d. ε_i with $\text{var}(\varepsilon_i) = \sigma^2$, and F_X the distribution of X . *Prediction risk*,

$$\begin{aligned}\mathbb{E}[(Y - \hat{m}(X))^2] &= \sigma^2 + \mathbb{E} \int (m(x) - \hat{m}(x))^2 dF_X(x) \\ &= \sigma^2 + \int \text{bias}(\hat{m}(x))^2 + V(\hat{m}(x)) dF_X(x) = \sigma^2 + \|m - \hat{m}\|_2^2\end{aligned}$$

- *Universal consistency*: if $\|\hat{m} - m\|_2 \xrightarrow{P} 0$, then $\hat{m}$ is consistent

Motivation - k -nearest neighbour

- To motivate, start with a nearest neighbour estimator $\hat{m}$.
- Define, $\mathcal{N}_k(x)$ as the indices, i , of the k closest points of $X_1, \dots, X_n$ to x ,

$$\hat{m}(x) = \frac{1}{k} \sum_{i \in \mathcal{N}_k(x)} Y_i = \sum_{i=1}^n w_i(x) Y_i$$

- Very “jagged” shape: $w_i(x) = 1/k$ if $i \in \mathcal{N}_k(x)$, 0 otherwise
- *Linear smoother*: $\hat{\mu} = (\hat{m}(X_1), \dots, \hat{m}(X_n))$, then $\hat{\mu} = SY$ for fixed inputs $(X_1, \dots, X_n)$
- *Universal consistency*: $\mathbb{E}\|m - \hat{m}\|_2^2 \rightarrow 0$ under only $\mathbb{E}Y_i^2 < \infty$ if $k = k_n \rightarrow \infty$, $k_n/n \rightarrow 0$
[Györfi, Kohler, Krzyzak and Walk (2006)]

Motivation - k -nearest neighbour

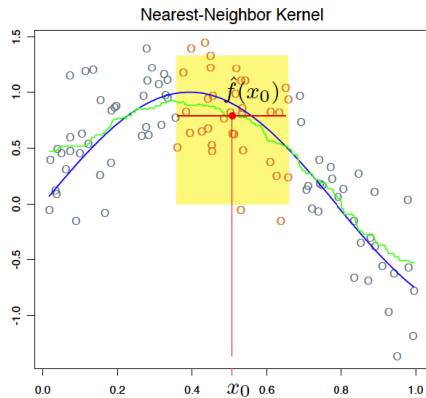


Figure: [Hastie, Tibshirani and Wainwright (2015)]

Motivation - k -nearest neighbour

- [Györfi, Kohler, Krzyzak and Walk (2006)] show for $k = k_n \rightarrow \infty$, $k_n/n \rightarrow 0$,

$$\mathbb{E}\|m - \hat{m}\|_2^2 \lesssim n^{-2/(2+d)}$$

under smoothness of m , specifically, Lipschitz $|m(x) - m(x')| \leq L\|x - x'\|$

- Sketch: (for simplicity, fix inputs, i.e. condition on X_i)

$$\begin{aligned}\mathbb{E}[(m(x) - \hat{m}(x))^2] &= \mathbb{E}[(m(x) - \mathbb{E}[\hat{m}(x)] + \mathbb{E}[\hat{m}(x)] - \hat{m}(x))^2] \\ &= (m(x) - \mathbb{E}[\hat{m}(x)])^2 + \mathbb{E}[(\hat{m}(x) - \mathbb{E}[\hat{m}(x)])^2] \quad \left\{ = \text{Bias}(\hat{m}(x))^2 + \text{Var}(\hat{m}(x)) \right\} \\ &= \left(\frac{1}{k} \sum_{i \in \mathcal{N}_k(x)} (m(X_i) - m(x)) \right)^2 + \frac{\sigma^2}{k} \\ &\leq \left(\frac{L}{k} \sum_{i \in \mathcal{N}_k(x)} \|X_i - x\| \right)^2 + \frac{\sigma^2}{k}\end{aligned}$$

- Bias-variance trade-off: expect first term to decrease with k , second term increases

Motivation - k -nearest neighbour

- To complete analysis, need some understanding of distribution of X_i
- Expect for “most” points x , $\|X_i - x\| \leq C(k/n)^{1/d}$
- Think about a line vs. 2D grid space with X_i uniformly distributed over unit ball, i.e. for sample size n , $X_i = i/n$.
- Line: $\max_{i \in \mathcal{N}_k(x)} |X_i - x| = k/n$. For $|X_i - x| \leq \epsilon/2$, $X_i = i/n \in [x - \epsilon/2, x + \epsilon/2]$. $P_X([x - \epsilon/2, x + \epsilon/2]) = \epsilon$, hence $k = n\epsilon$ such that $\epsilon = k/n$.
- Grid: $\max_{i \in \mathcal{N}_k(x)} \|X_i - x\| \lesssim C(k/n)^{1/2}$. For $\|X_i - x\| \leq \epsilon/2$, $X_i \in [x - \epsilon/2, x + \epsilon/2]^2$. Then $P_X([x - \epsilon/2, x + \epsilon/2]^2) = \epsilon^2$, and $k = n\epsilon^2$ such that $\epsilon = (k/n)^{1/2}$
- Argument extends to d -dimensions (and more arbitrary X_i domains). Then,

$$\mathbb{E}[(m(x) - \hat{m}(x))^2] = (LC)^2 \left(\frac{k^*}{n}\right)^{2/d} + \frac{\sigma^2}{k^*} = n^{-\frac{2}{2+d}}$$

where $k^* = n^{2/(2+d)}$ sets order of first and second term equal

Kernel Regression

- Consider kernel functions $K : \mathbb{R} \rightarrow \mathbb{R}$,

$$\int K(u) du = 1, \quad \int uK(u) du = 0, \quad \mu_2(K) := \int u^2 K(u) du < \infty.$$

- Common examples (box-car, Gaussian, Epanechnikov):

$$K(u) = \mathbb{1}\{u \leq 1/2\}, \quad K(u) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{u^2}{2}\right), \quad K(u) = \mathbb{1}\{u \leq 1\}(3/4)(1 - u^2)$$

- Denote $K_h(X_i - x) := h^{-1}K(\|X_i - x\|/h)$. Nadaraya-Watson estimator (a linear smoother)

$$\hat{m}(x) = \frac{\sum_i K(\|X_i - x\|/h) Y_i}{\sum_i K(\|X_i - x\|/h)} = \frac{\sum_i K_h(X_i - x) Y_i}{\sum_i K_h(X_i - x)} = \sum_i w_i(x) Y_i$$

Kernel Regression

- To derive the Nadaraya-Watson estimator, consider

$$\hat{f}(x) = n^{-1} \sum_i K_h(X_i - x); \quad \hat{f}(x, y) = n^{-1} \sum_i K_h(X_i - x) K_h(Y_i - y)$$

- Use the definition for conditional expectations,

$$\mathbb{E}[Y|X = x] = \int y f(y|x) dy = \int y f(y, x) / f(x) dy$$

- Then, for $\hat{\mathbb{E}}[Y|X = x]$ replace $f(y, x)/f(x)$ with $\hat{f}(y, x)/\hat{f}(x)$,

$$\hat{\mathbb{E}}[Y|X = x] = \frac{\sum_i K_h(X_i - x) \int y K_h(Y_i - y) dy}{\sum_i K_h(X_i - x)} = \frac{\sum_i K_h(X_i - x) Y_i}{\sum_i K_h(X_i - x)}$$

Kernel Regression

- Now to study MSE, consider scalar X_i ,
- For $\hat{r}(x) := \hat{m}(x)\hat{f}(x)$ have, $\hat{m}(x) = \hat{r}(x)/\hat{f}(x)$. Then,

$$\begin{aligned}\mathbb{E}\hat{r}(x) &= n^{-1}\mathbb{E}\sum_i K_h(X_i - x)Y_i = \int \int K_h(x' - x)f(y, x')dydx' \\ &= \int K_h(x' - x)\left(\int f(y|x')dy\right)f(x')dx' = \int K_h(x' - x)m(x')f(x')dx'\end{aligned}$$

- For $r(x') := m(x')f(x')$, $u = (x' - x)/h$ such that $dx' = h \cdot du$

$$\begin{aligned}\int K_h(x' - x)m(x')p(x')dx' &= \int K(u)r(uh + x)du \\ &= r(x) \int K(u)du + hr'(x) \int uK(u)du + h^2r''(x) \int u^2K(u)du + o(h^2) \\ &= r(x) + h^2r''(x)\mu_2(K) + o(h^2)\end{aligned}$$

Kernel Regression

- For variance, define $\sigma^2(x) := \mathbb{E}[Y^2|X = x]$,

$$\begin{aligned}\text{Var}(\hat{r}(x)) &= \text{Var}\left(n^{-1} \sum_i K_h(X_i - x) Y_i\right) = n^{-1} \text{Var}(K_h(X_i - x) Y_i) \\ &= n^{-1} \left[\int K_h^2(x' - x) \left(\int y^2 f(y|x') dy \right) f(x') dx' - \left(\int K_h(x' - x) m(x') f(x') dx' \right)^2 \right] \\ &= n^{-1} \left[\int K_h^2(x' - x) \sigma^2(x') f(x') dx' - \left(\int K_h(x' - x) m(x') f(x') dx' \right)^2 \right]\end{aligned}$$

- Use $u := (x' - x)/h$, then, $\sigma^2(x + uh)f(x + uh) = \sigma^2(x)f(x) + O(h)$, and,

$$\text{Var}(\hat{r}(x)) = n^{-1} \int K_h^2(x' - x) \sigma^2(x') f(x') dx' + O(n^{-1}) = (nh)^{-1} \sigma^2(x) f(x) \|K\|^2 + o((nh)^{-1})$$

where $\|K\|^2 = \int K^2(u) du$. Hence by Slutsky's Theorem $\hat{m}(x) \xrightarrow{P} m(x)$ if $h \rightarrow 0$, $nh \rightarrow \infty$

Kernel Regression – bandwidth selection

- Here we explore some high-level asymptotic rates for bandwidth selection
- Many finite sample considerations should be made, we ignore here
- How to optimise MSE with respect to h ?

$$\hat{m}(x) - m(x) = \frac{\hat{r} - m\hat{f}}{\hat{f}} = \frac{\hat{r} - m\hat{f}}{\hat{f}} \left(\frac{\hat{f}}{f} - \frac{f - \hat{f}}{f} \right) = \frac{\hat{r} - m\hat{f}}{f} + (\hat{m} - m) \frac{f - \hat{f}}{f}$$

- The second term is $o_p(1)O_p(h^2 + (nh)^{-1/2})$ by consistency of $\hat{m}$, and consistency of $\hat{f}$
- Numerator of first term is $r + O_p(h^2) + O((nh)^{-1/2}) - m(f + O_p(h^2 + (nh)^{-1/2}))$ by consistency of $\hat{r}$ and $\hat{f}$

$$MSE(h) = O(h^4) + O((nh)^{-1})$$

- To equalise terms, set $h^* = n^{-1/5}$, which leads to $MSE(h^*) = n^{-4/5}$

Kernel Regression – bandwidth selection

- For scalar X_i MSE bounded in asymptotic order,

$$MSE(h) = O(h^4) + O((nh)^{-1})$$

- To equalise terms, set $h^* = n^{-1/5}$, which leads to $MSE(h^*) = O(n^{-4/5})$
- For vector valued $X_i \in \mathbb{R}^d$,

$$MSE(h) = O(h^4) + O((nh^d)^{-1})$$

i.e. higher variance since effective sample size scales exponentially (downwardly) in d

- Optimal $h^* = n^{-1/(4+d)}$ for vector case, with $MSE(h^*) = O(n^{-4/(4+d)})$

Kernel Regression – bandwidth selection w cross validation

- Cross-validation is a useful tool for hyperparameter optimisation
- Bandwidth here is the hyperparameter – i.e. not defined by data generating process
- Cross-validation uses part of the data to train the model, then another part to fit the model and measure empirical MSE just from the fitted section of the data
- Leave-One-Out-Cross-Validation (LOOCV): let $\hat{m}_{-i,h}(x)$ be the estimator $\hat{m}(x)$ without the i -th observation
- Observe $Y_i = m(X_i) + \epsilon_i$, and error $\hat{\epsilon}_{-i} = Y_i - \hat{m}_{-i,h}(x)$

$$\widehat{MSE}(h)_{LOOCV} = n^{-1} \sum_i (Y_i - \hat{m}_{-i,h}(x))^2$$

- Choose a grid of possible h (one-dimensional) then pick smallest $\widehat{MSE}(h)_{LOOCV}$
- Computationally simple for reasonably smooth function since h is one-dimensional
- Can extend to leave- k out

References

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